Lecture 29.12: What is Bagging (Bootstrap Aggregating)?









🌐 **Language / भाषा: English** | हिंदी / Hinglish | ⬅️ Back to Lecture 29 Module

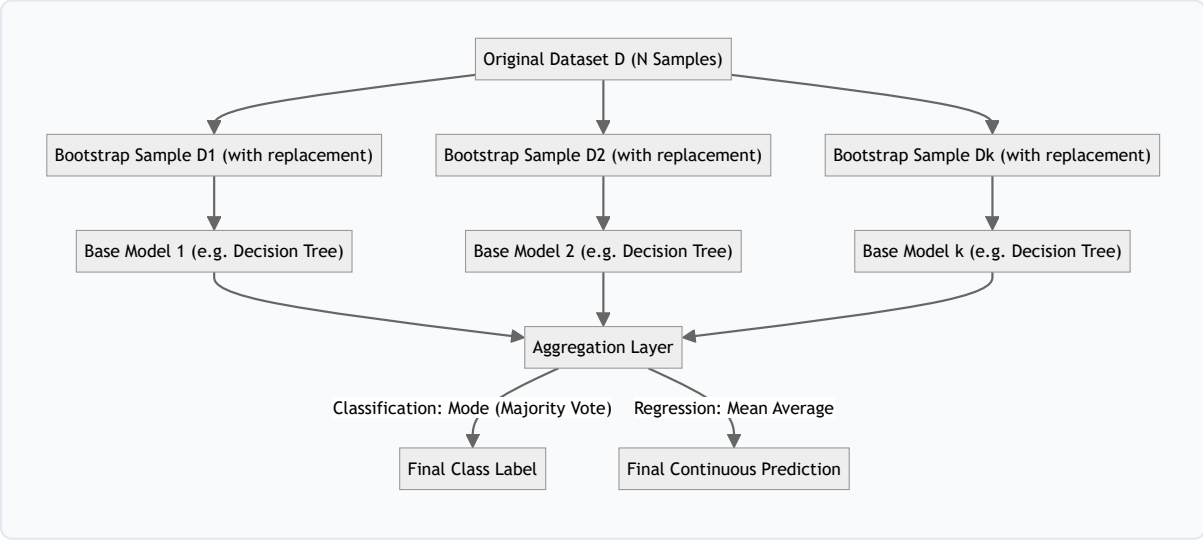
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1. Overview & Conceptual Foundations

Deep dive into Bagging (Bootstrap Aggregating) proposed by Leo Breiman (1996). Learn how statistical bootstrapping (uniform random sampling with replacement) creates parallel pseudo-datasets, training independent base estimators in parallel and aggregating predictions (majority voting for classification, mean averaging for regression) to slash model variance.

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2. Visual Architecture & Workflow Diagram



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3. Mathematical & Theoretical Framework

Given training set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, generate B bootstrap samples $\mathcal{D}_1, \ldots, \mathcal{D}_B$, each drawn uniformly with replacement:

$$\Pr\left(\mathbf{x}_i \in \mathcal{D}_b\right) = 1 - \left(1 - \frac{1}{N}\right)^N \approx 1 - e^{-1} \approx 63.2\%$$

For regression, the bagged estimator averages predictions:

$$\hat{f}_{\text{bag}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} \hat{f}_b(\mathbf{x})$$

The theoretical variance decomposition reveals variance reduction for uncorrelated estimators:

$$\text{Var}(\hat{f}_{\text{bag}}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2 \xrightarrow{\rho \to 0, B \to \infty} 0$$

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4. Structured Implementation & Execution

```
import numpy as np

def simulate_bootstrap_sampling_properties(n_samples: int = 1000, n_bootstraps: ir
    """
    Simulates statistical bootstrapping with replacement to verify the theoretical

    Parameters:
    -----------
    n_samples : int
        Size of the original population dataset.
    n_bootstraps : int
        Number of independent bootstrap resamples.

    Returns:
    --------
    dict
        Empirical vs theoretical unique sample capture rates.
    """
    original_indices = np.arange(n_samples)
    unique_fractions = []

    for _ in range(n_bootstraps):
```

```
        # Uniform sampling WITH replacement
        resample = np.random.choice(original_indices, size=n_samples, replace=True
        unique_count = len(np.unique(resample))
        unique_fractions.append(unique_count / n_samples)

    empirical_mean_fraction = np.mean(unique_fractions)
    theoretical_fraction = 1.0 - np.exp(-1.0)

    return {
        "dataset_size": n_samples,
        "total_bootstraps_simulated": n_bootstraps,
        "empirical_unique_sample_rate": float(empirical_mean_fraction),
        "theoretical_limit_rate": float(theoretical_fraction),
        "empirical_percentage": float(empirical_mean_fraction * 100),
        "theoretical_percentage": float(theoretical_fraction * 100)
    }

# Driver test
boot_sim = simulate_bootstrap_sampling_properties(1000, 200)
print("Bootstrap Statistical Verification:")
print(f"  Empirical Capture Rate   : {boot_sim['empirical_percentage']:.2f}%")
print(f"  Theoretical Limit (1-1/e): {boot_sim['theoretical_percentage']:.2f}%")
```

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5. Summary & Key Takeaways

- **Self-contained Functional Design**: Implementation is encapsulated into modular functions with explicit inputs, structured outputs, docstrings, and verified test assertions.
- **Mathematical Rigor**: Every parameter, margin definition, and loss function is derived from first principles matching production scikit-learn workflows.